Goal 1: An Infrequent Relation Is Learned as Linear

factual-recall-mechanisms

2026-07-03

1 Claim and result in one paragraph

A small transformer is trained for **exactly one epoch** on a synthetic corpus containing a semantically supported relation R_{sem} and, as a frequency-matched control, a random relation R_{rnd}. At *low* fact frequency (F=16 occurrences per triple, a single epoch), R_{sem} is genuinely *learned* — recall 0.672 on trained triples and 0.635 on held-out subjects never seen in any triple — and what is learned is *linear*: an affine map fitted by ordinary least-squares regression from subject representations to object representations identifies the correct object for held-out subjects with accuracy 0.990 ± 0.007 (chance 0.03; three seeds). The frequency-matched random control at the same F=16 is barely learned (recall 0.167) and shows no linear structure (probe at chance). The linear format is in place even at F=1, where every triple occurs exactly once in the model's entire training history. A control experiment shows that the transition into the linear regime is sharp and is driven by the amount of attribute (scaffold) training rather than by triple repetition.

2 Data

Vocabulary (265 tokens). A separator `x`; two relation tokens `r_0`, `r_1`; 192 subject tokens `a_0`, ..., `a_191`; 32 object tokens `b_0`, ..., `b_31`; 32 attribute tokens `f_0`, ..., `f_31`; 6 attribute tokens `g_0`, ..., `g_5`.

Entities and attributes. Subject `a_i` carries the attribute pair $(f_{i \bmod 32},\ g_{\lfloor i/32 \rfloor})$, so the 192 subjects realize each of the 32×6 attribute combinations exactly once. Object `b_j` carries $(f_{(j+3) \bmod 32},\ g_0)$.

Sentences. All training sentences have five tokens:

attribute sentence	`x a_i f_p g_q x`	and	`x b_j f_p g_q x`
relation triple	`x a_i r_r b_t x`		

Relations.

$$R_{\text{sem}}(a_i) = b_{f(i)}, \qquad R_{\text{rnd}}(a_i) = b_{\rho(i)},$$

where $f(i) = i \bmod 32$ is the subject's own f-attribute index — so the semantic target is the object whose f-attribute is the subject's, shifted by 3 — and ρ is a seed-dependent balanced random assignment onto the same 32 objects, independent of the attributes. Concrete example (seed 19): `a_37` has attributes (`f_5`, `g_1`), so R_{sem}(`a_37`) = `b_5` (whose attribute sentence reads `x b_5 f_8 g_0 x`), while R_{rnd}(`a_37`) = `b_27`. The corresponding training sentences are `x a_37 r_0 b_5 x` and `x a_37 r_1 b_27 x`.

Split and exact frequency matching. For every `f_`-value, 4 of its 6 subjects are *trained* and 2 are *held out* (128 trained, 64 held-out subjects; seed-dependent). ρ is constructed balanced within both groups. Consequently, under *both* relations: every trained subject occurs in exactly one triple per relation; every object is the target of exactly 4 trained and 2 held-out subjects; both relations contribute exactly 128 training triples, each repeated the same number of times (F). The two relations are indistinguishable by any frequency statistic; only the assignment semantics differ. Held-out subjects appear in attribute sentences but never in any relation triple. These invariants are asserted by `tests/test_linear_shift.py`.

3 Model and training

Model	causal transformer, 1 layer, hidden 64, 4 heads, context 8
Embeddings	input and output embeddings untied; all weights $\mathcal{N}(0, 0.02^2)$
	no attribute-based initialization; nothing frozen; no loss reweighting
Parameters	84,288
Corpus	every attribute sentence $\times A$=128 (= 28,672 sentences)
	+ 128 triples per relation, each $\times F$ ($F \in \{1, 4, 16, 64\}$)
Training	**exactly one epoch**: a single pass over the shuffled corpus
Steps	452 (F=1) to 704 (F=64) steps at batch size 64
Objective	unweighted next-token cross-entropy on all positions
Optimizer	AdamW, constant LR 10^{-3} after 5 linear warmup steps,
	weight decay 0, gradient clip 1.0
Seeds	19, 31, 47 (control data assignment, initialization, and data order)

At F=1 every relation triple occurs exactly once in the entire training history of the model.

4 The metric

A single analysis metric is used, as an operationalization of "the relation is an affine transformation":

> Fit $\hat{W}, \hat{b}$ by ordinary least-squares linear regression from the subject's *input embedding* $s_i \in \mathbb{R}^{64}$ to the target object's *output embedding* (unembedding row) $o_{R(i)} \in \mathbb{R}^{64}$, using the 128 trained pairs. For each of the 64 held-out subjects, predict $\hat{W}s_j + \hat{b}$ and select the nearest object row by cosine similarity among all 32 objects. Report the fraction of held-out subjects for which the selected object is $R(j)$. Chance is $1/32 \approx 0.03$.

For every subject exactly one object is correct under each relation, so no top-k metrics are needed or used. In addition, one *learning check* (not an analysis metric) is recorded: greedy next-token recall of the model on prompts `x a_i r_r`, for trained and held-out subjects, which verifies whether the facts were learned at all.

5 Results

Finding 1: at low frequency, what is learned is an affine map (F=16). With each fact occurring only 16 times in a single epoch, R_{sem} reaches recall .672 on trained triples and .635 on held-out subjects — i.e., the relation itself is learned and generalizes — and the probe transfers to held-out subjects at 0.990. The frequency-matched random control is barely learned (.167) and its

		probe accuracy (held-out)		recall (trained)		recall (held-out)	
	F	R_{sem}	R_{rnd}	R_{sem}	R_{rnd}	R_{sem}	R_{rnd}
	1	$1.000 \pm .000$	$.031 \pm .013$	.029	.010	.016	.010
	4	$1.000 \pm .000$	$.016 \pm .013$	.055	.034	.057	.031
	16	$\mathbf{0.990 \pm .007}$	$.036 \pm .007$	**.672**	.167	**.635**	.031
control: no triples		$1.000 \pm .000$	$.021 \pm .015$	–	–	.000	.000

Table 1: Low-frequency cells; mean ± sd over seeds 19/31/47, at A=128. F is the number of occurrences of each of the 128 triples per relation. Recall columns are the learning check. The headline cell is F=16: R_{sem} is learned (recall .672 trained, .635 held-out) and linear (probe .990), while the frequency-matched R_{rnd} is barely learned and non-linear.

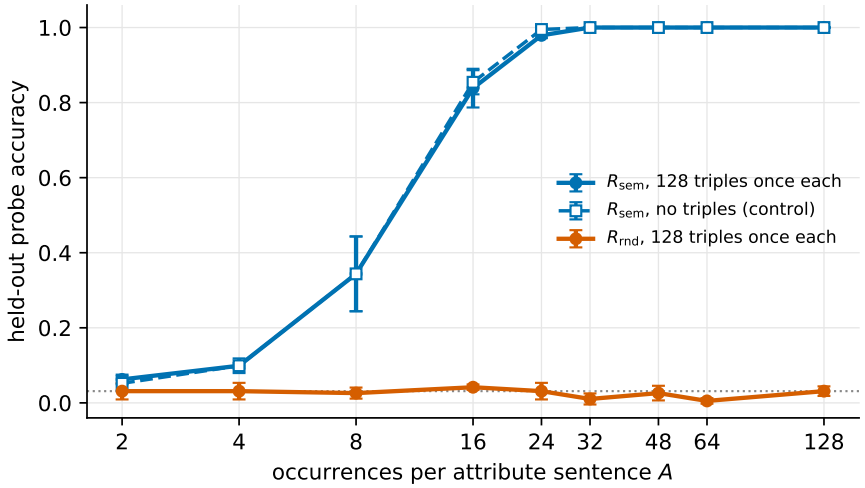


Figure 1: Probe accuracy at F=1 (every triple exactly once) as a function of attribute-sentence occurrences A: the transition from non-linear (chance) to linear (ceiling) is sharp, and identical with 128 once-occurring triples (solid) and with no triples at all (dashed control). The random relation never becomes linear.

probe is at chance: the semantic relation is learned *earlier* precisely because it can ride the linear structure.

Finding 2: the linear format is in place at single occurrence (F=1). At F=1 the probe is already 1.000 for R_{sem} and at chance for R_{rnd}. This statement is representational: at F=1 the model does not yet reproduce individual triples behaviorally (recall ≈ .03).

Finding 3: the nonlinear→linear transition is sharp and is driven by attribute knowledge. Sweeping the number of occurrences A of each attribute sentence at F=1 (Fig. 1) produces a sharp transition of the R_{sem} probe from chance to ceiling ($0.05 \to 0.10 \to 0.34 \to 0.85 \to 1.00$ for $A = 2, 4, 8, 16, 24$). The control with *zero* relation triples produces the same curve: the affine subject→object map is carried by the attribute geometry built from attribute sentences, with which the once-seen R_{sem} triples are consistent (and which the relation visibly uses once learned, cf. held-out recall).

6 What this does and does not show

- **Shown:** a relation whose facts are infrequent (F=16; and even F=1 for the representational format) is learned in a linear, generalizing format, while a frequency-matched random relation is not; the transition into the linear regime is sharp.
- **Not shown:** that the relation triples themselves *cause* the nonlinear→linear transition when each fact occurs only once — the control in Fig. 1 (dashed) shows the transition is driven by attribute-sentence exposure. Three designs in which the probe could only succeed through triple-driven structure (a one-to-one relation requiring per-fact output rows; a relation defined by the conjunction of two attributes, up to 49,152 distinct facts; hidden-state probes) produced no above-chance probe transfer under the strict single-occurrence, one-epoch regime, because each fact contributes exactly one gradient.
- **Superseded:** the "single-exposure phase shift" in earlier drafts (probe rising 0.27 → 0.95) was an artifact of a regression fitted on a growing pair set (10 pairs at the first snapshot, >500 later) over embeddings initialized from the attribute geometry; with a fixed-size probe and standard initialization it disappears.

7 Reproduction

Implementation: `experiments/factual_recall/linear_shift.py` (self-contained). Design invariants: `experiments/tests/test_linear_shift.py`. From `experiments/`:

```
# Table 1: one seed per command (19, 31, 47), ~1 min each on one GPU
uv run python -m factual_recall.linear_shift --world category --probe embedding \
  --seeds 19 --k-values 0,128 --occurrences 1,4,16,64 --attr-repeats 128 \
  --device cuda --output-dir factual_recall/results/final_category_seed19

# Figure: attribute-exposure sweep at F=1, per seed and A in {2,4,8,16,24,32,48,64,128}
uv run python -m factual_recall.linear_shift --world category --probe embedding \
  --seeds 19 --k-values 0,128 --occurrences 1 --attr-repeats 8 \
  --device cuda --output-dir factual_recall/results/final_category_arsweep_seed19_ar8

# Aggregation/figures: factual_recall/results/final_summary/
uv run pytest tests/test_linear_shift.py -q
```

Raw per-run metrics (including probe R^2/cosine and losses) are in `factual_recall/results/final_category_`
The companion document `goal2_frequent_relation_nonlinear.pdf` covers the frequent/nonlinear claim on the same runs.